



Algal species as substrates for cultivating potentially pathogenic free-living amoebae: a novel in vitro strategy

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Abstract

Free-living amoebae (FLA) are ubiquitous protists of ecological and clinical importance whose in vitro cultivation traditionally relies on bacterial food sources, most commonly *Escherichia coli* on non-nutrient agar. Although this approach is widely used, bacteria-based substrates may introduce experimental and biosafety-related limitations because some bacteria can survive phagocytosis and persist within amoebae, which may act as intracellular reservoirs that protect endocytobionts from environmental stress and antimicrobial treatments and may also promote bacterial interactions associated with horizontal gene transfer and the dissemination of virulence- or resistance-related traits. To evaluate a non-bacterial alternative for laboratory cultivation, this study assessed three green algal species — *Scenedesmus obliquus*, *Chlamydomonas reinhardtii*, and *Chlorella vulgaris* — as trophic substrates for three potentially pathogenic FLA on non-nutrient agar, using *E. coli* as the conventional positive control. Among the tested amoebae, *Acanthamoeba* genotype T3 (accession no. OR568985) exhibited the greatest growth across all tested substrates, with the highest cell densities recorded on *S. obliquus*, followed by *C. reinhardtii* and *E. coli* by day 7. *Naegleria clarki* (accession no. OR568992) and *Vermamoeba vermiformis* (accession no. OR568993) also proliferated on algal substrates, although growth was generally higher on *E. coli*. Strong positive correlations between algal- and bacterial-based growth patterns indicated broadly similar temporal trends in amoebal proliferation. Overall, these findings demonstrate that selected green algae, particularly *S. obliquus* and *C. reinhardtii*, can support measurable growth of potentially pathogenic FLA and represent promising non-bacterial substrates for laboratory cultivation.

Keywords Algae · Alternative food source · In vitro · *Acanthamoeba* genotype T3 · *Naegleria clarki* · *Vermamoeba vermiformis*

Introduction

Free-living amoebae (FLA) are ubiquitous microorganisms distributed across a wide range of natural and engineered environments, including soil, freshwater, wastewater, and air (Fonseca et al. 2020; Fan et al. 2024; Gad et al. 2024a,

b; Ghadesi et al. 2024; Shaheen et al. 2024). They contribute to microbial grazing and nutrient turnover and thus play important ecological roles in regulating microbial community structure (Delafont et al. 2016; Archibald et al. 2017). Despite their ecological significance, FLA are also recognized as opportunistic pathogens capable of causing severe

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infections in humans and animals, particularly among immunocompromised individuals (Marciano-Cabral and Cabral 2003; Król-Turmińska and Olender 2017). For instance, species belonging to genera *Acanthamoeba*, *Naegleria*, and *Vermamoeba* are implicated in human infections. *Acanthamoeba* species are associated with a range of clinical manifestations, including *Acanthamoeba* keratitis (AK), granulomatous amoebic encephalitis (GAE), and disseminated infections in immunocompromised hosts (Marciano-Cabral and Cabral 2003; Visvesvara et al. 2007; Belaidi et al. 2025).

The public health importance of FLA extends beyond their direct pathogenicity. Numerous studies have shown that bacteria engulfed by amoebae may survive intracellular digestion and persist as endocytobionts (Mungroo et al. 2021). In this protected intracellular niche, amoebae may function as environmental reservoirs or “Trojan horses,” shielding internalized bacteria from harsh conditions such as desiccation, disinfectants, antibiotics, and other biocidal treatments (Barker et al. 1999; Scheid 2014). Moreover, residence within amoebae may enhance bacterial persistence, promote virulence-related adaptation, and create conditions favorable for interactions associated with horizontal gene transfer and the dissemination of resistance- or virulence-related traits (Mungroo et al. 2021). Accordingly, the use of bacterial prey in laboratory cultivation systems may introduce biological and interpretive complexities that extend beyond a simple nutritional role.

Prey-cell-free cultivation of FLA is also possible using enriched axenic media. PYG is well established for many adapted *Acanthamoeba* strains, whereas *Naegleria*, *Vermamoeba*, and other FLA may require different supplemented formulations. Because nutritional requirements and adaptation to axenic growth vary among genera and strains, no single prey-free medium can be assumed to support diverse FLA taxa equally (Schuster 2002). The present study therefore evaluated green algal cells as non-bacterial particulate substrates in an agar-based cultivation system comparable with the conventional bacterial-food approach.

Despite the recognized ecological and clinical importance of FLA, culture-based methods remain fundamental for their isolation, maintenance, and experimental study. Among available approaches, non-nutrient agar (NNA) seeded with bacteria, particularly *E. coli*, is widely used as the conventional cultivation method (Khan 2006). This system is practical, effective, and well established, and *E. coli* therefore represents an appropriate positive control for comparative cultivation studies. However, although bacteria-based feeding systems are highly useful for routine amoeba growth, they may not always be optimal when a more controlled (Martinez 2019; Mungroo et al. 2022), non-bacterial trophic substrate is desired, especially in experiments where intracellular bacteria-associated interactions may confound downstream observations or biosafety considerations.

In this context algae represent an attractive alternative trophic resource. As photosynthetic eukaryotes, algae are phylogenetically distinct from bacteria and may provide an alternative non-bacterial food source for amoeba cultivation, potentially reducing experimental and biosafety concerns associated with bacteria–amoeba interactions, including intracellular bacterial persistence and adaptation (Van Wichelen et al. 2010; Mungroo et al. 2021; Weger et al. 2024). In addition, replacing bacterial prey with algal substrates may reduce the likelihood that cultivation systems unintentionally support amoeba-associated bacterial persistence or bacteria-mediated interactions linked to intracellular survival (Mungroo et al. 2021). Previous ecological studies have shown that protozoa can exploit algal resources under various environmental conditions (Boenigk and Arndt 2002; Van Wichelen et al. 2010; Weger et al. 2024); however, the use of algae as intentional non-bacterial substrates for cultivating potentially pathogenic FLA on NNA has remained insufficiently explored.

Previous studies have demonstrated that certain amoebae can utilize selected green algae as food sources; however, successful growth appears to depend on the specific amoeba–alga combination. Early cultivation experiments showed considerable variation in the ability of different algal species to support amoeba growth (Ho and Alexander 1974). More recently, Weger et al. (2024) tested several green algae against three amoebae and found that only one combination, *Cochliopodium* sp. with *Scenedesmus vacuolatus*, supported substantial growth. These observations indicate that green algae should not be considered universally suitable substrates for amoebae and highlight the need to evaluate defined algal species against individual FLA taxa under standardized cultivation conditions.

Therefore, the present study evaluated three green algal species, *Scenedesmus obliquus*, *Chlamydomonas reinhardtii*, and *Chlorella vulgaris*, as alternative trophic substrates for the cultivation of three potentially pathogenic FLA, namely *Acanthamoeba* genotype T3, *Naegleria clarki*, and *Vermamoeba vermiformis*, using *E. coli* as the conventional positive control. We hypothesized that selected algal substrates could support measurable amoebal growth and thus provide promising non-bacterial alternatives for laboratory cultivation.

Materials and methods

The source of FLA, bacteria, and algae strains

Three FLA species, *Acanthamoeba* genotype T3 (accession no. OR568985), *Naegleria clarki* (OR568992), and *Vermamoeba vermiformis* (OR568993), were isolated and identified from the outlets of reverse osmosis point-of-use

systems (Gad et al. 2024a). Sequencing of the isolates was performed using the QPinnacle2 platform (QitanTech, China). These FLA species were selected for use in the present work as they are globally and locally dominant taxa in natural and engineered freshwater systems. Moreover, they have documented clinical relevance, and are known to harbor and interact with a range of intracellular pathogenic bacteria under environmental and in vitro conditions (Trabelsi et al. 2012; Salahuldeen et al. 2020; Gad et al. 2024b). These FLA species were used in subsequent trials with monoxenic media to explore a novel food source. Professor Gamila Hussien Ali kindly provided the algal strains from the National Laboratory of Algae at the National Research Centre (Egypt). *Escherichia coli* (ATCC 25922) bacteria were kindly provided by Professor Gamila El-Taweel, Environmental Microbiology Laboratory, National Research Centre, Egypt.

Preparation of FLA cysts

The previously isolated and molecularly identified strains of FLA (Gad et al. 2024a) were separately propagated on NNA seeded with heat killed *E. coli* and allowed to encyst at room temperature. The cysts were scraped from the agar surface, transferred into tubes, and washed five times with sterile Page's Amoebae Saline (PAS) to get rid of suspending bacteria in the supernatant. After each wash, the tubes containing cysts of amoebae were centrifuged at 1500 rpm for 10 min and the supernatant was discarded. The pellet, containing amoebae cysts, was then re-suspended in 1 mL of sterile PAS.

To ensure the absence of bacteria after the last wash, the obtained cysts were cultivated on non-nutrient agar without any food source. If cultured amoebic cysts did not transform to active trophozoites, this is an indication to ensure the complete absence of bacteria from washed cysts (i.e., washed amoebic cysts became free from bacteria that could interfere with the inoculated food sources).

Counting of the three different strains of washed amoebic cysts was carefully performed by three independent researchers. Counted cysts were dispensed in Eppendorf tubes (each having a mean count of 10 ± 2 cysts/tube) to be ready for use in subsequent experiments.

Use of algal species for FLA cultivation

Three algal species, *Scenedesmus obliquus*, *Chlamydomonas reinhardtii*, and *Chlorella vulgaris* were separately tested as alternative food sources to replace *E. coli*, which is conventionally used for culturing FLA (Salahuldeen et al. 2020; Gad et al. 2024a). The selected algal taxa represent common freshwater chlorophytes frequently detected in aquatic systems. *Chlamydomonas reinhardtii* is a well-established

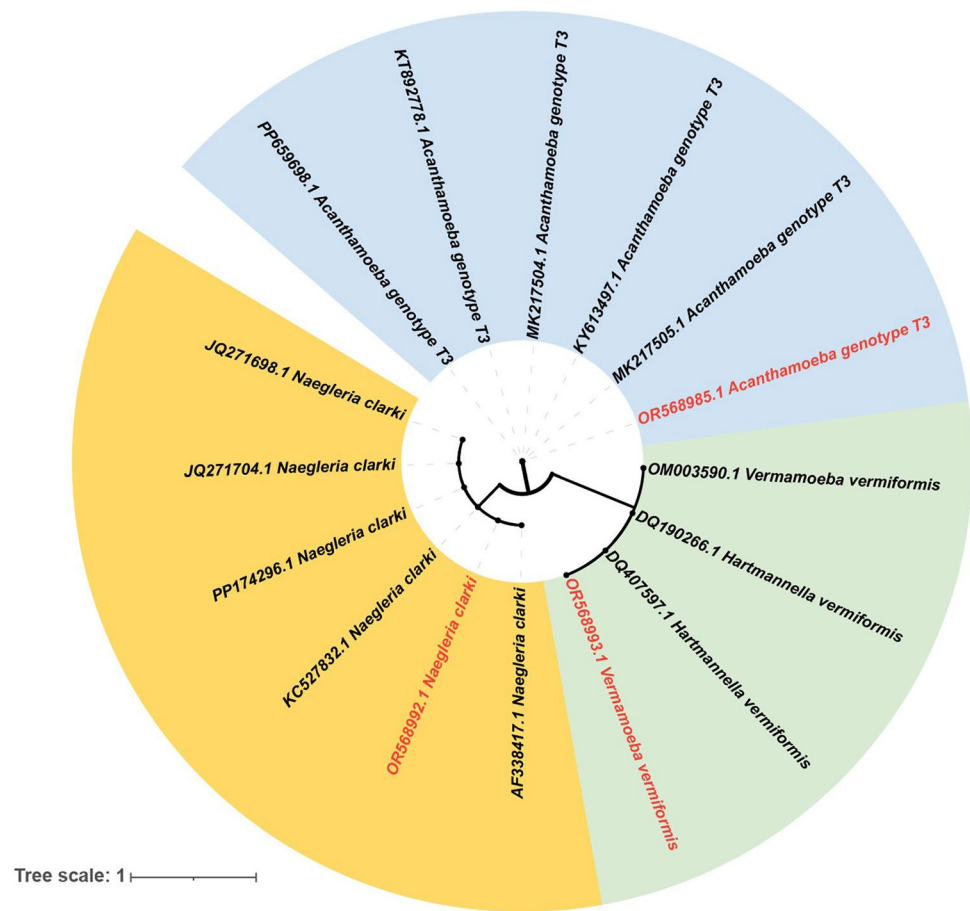
laboratory model organism with extensive physiological characterization; *S. obliquus* is a widespread colonial green algal species commonly present in biofilms and surface waters; and *C. vulgaris* is widely used in ecological and applied research because of its robust growth and ease of cultivation and is characterized by broad environmental distribution and tolerance to environmental stress. These algal species differ in morphology, cell size, and cell wall composition, allowing assessment of potential species-specific growth responses of FLA (Hodač et al. 2016; Lamminen 2021; Oliveira et al. 2021; Calatrava et al. 2023). Previous ecological studies have demonstrated that protists, including amoebae and other microeukaryotes, can utilize green algae as food sources (Martinez 1996; Schuster 2002; Wang et al. 2014), supporting their relevance for evaluating algal-based cultivation approaches.

Non-nutrient agar (NNA) in sterile 24-well plates was seeded with either heat-killed *E. coli* (forming a bacterial lawn) or algal suspensions. For bacterial controls, *E. coli* was applied to NNA to support trophozoite growth and incubated at 37 °C following established protocols (Schuster 2002; Al-Herrawy and Gad 2015). The algal strains were maintained in BG-11 medium under controlled laboratory conditions at 25 ± 2 °C under continuous illumination until they reached the exponential growth phase (Andersen and Kawachi 2005). Algal cultures were washed three times with sterile distilled water prior to use as food source for FLA to reduce potential bacterial contamination. Each well was inoculated with 10 cysts, as described above. Experiments were conducted in triplicate wells per condition and *E. coli* controls were included in parallel.

Two substrate volumes (50 µL and 100 µL) from each species were separately tested to explore potential dose-dependent effects of food availability for FLA proliferation. These volumes corresponded to approximately 5×10^3 and 1×10^4 cells for algal suspensions and 2.1×10^4 and 1×10^6 cells for *E. coli*. Given the exploratory nature of this study, food substrates were standardized based on cell counts rather than biomass or carbon content. Because algal and bacterial cells differ in size and biochemical composition, strict nutritional equivalence was not established; therefore, the aim was to evaluate comparative growth responses under controlled experimental conditions rather than quantitative nutritional equivalence.

For growth assessment, three separate replicate wells were prepared for each experimental condition and sampling day. At each time point, the entire contents of each well were collected independently in 1 mL of sterile Page's Amoebae Saline (PAS), fixed with Lugol's iodine, and enumerated using a Neubauer hemocytometer (Koyun et al. 2020). All quantitative results are expressed as cells mL⁻¹ of the resuspended well contents. In this study, the term "cell" refers to either a trophozoite (T) or a cyst (C),

Fig. 1 Phylogenetic tree of the studied free-living amoeba sequences and related GenBank reference sequences. The circular maximum-likelihood tree was constructed from aligned nucleotide sequences and visualized using iTOL. Sequences from the present study are highlighted in red: *Naegleria clarki* OR568992.1, *Acanthamoeba* genotype T3 OR568985.1, and *Vermamoeba vermiformis* OR568993.1. Reference sequences are shown in black, with accession numbers preceding taxon names



as identified morphologically under light microscopy. Where indicated, total growth (T + C) represents the combined number of trophozoites and cysts per well, normalized to 1 mL.

Cell enumeration was conducted in an ISO/IEC 17025-accredited laboratory under documented standard operating procedures and periodic external evaluation. Each of the three replicate wells was independently evaluated by one of three skilled and experienced researchers trained in FLA morphological identification (Koyun et al. 2020). The reported values represent the mean $\pm$ standard deviation calculated from the three replicate wells ($n = 3$) for each experimental condition and sampling day. Separate sets of replicate wells were used for each sampling day, and growth was assessed from day 0 to day 7.

Statistical analysis

The differences in FLA growth on various food sources were assessed using Friedman, Kruskal–Wallis, and Mann–Whitney U tests. Additionally, Spearman correlation analysis was employed to evaluate the relationship between FLA growth on algal species and *E. coli*. All

statistical analyses and visualizations were conducted using R v4.1.0 (<https://www.r-project.org/>).

Results

Cultivation of FLA on non-nutrient agar seeded with *C. reinhardtii*

The phylogenetic analysis confirmed the taxonomic placement of the three studied free-living amoebae, with *N. clarki* OR568992.1, *Acanthamoeba* genotype T3 OR568985.1, and *V. vermiformis* OR568993.1 clustering within their corresponding GenBank reference groups (Fig. 1). The growth patterns of FLA revealed differences across species and food sources, highlighting variations in adaptability and nutrient preferences. Among the tested species, *Acanthamoeba* genotype T3 exhibited the highest overall growth, demonstrating its remarkable (Friedman test; $P < 0.0001$) adaptability to diverse substrates. When cultivated with *Chlamydomonas reinhardtii*, its growth increased exponentially from a mean of 10 on Day 0 to 2228 ± 47.52 cells mL^{-1} on Day 7 (Table S1, Fig. 3). In comparison, its growth means with *E. coli* reached 2171.6 ± 79.10 cells mL^{-1} on Day 7 (Table S2).

In contrast, *N. clarki* displayed a slightly lower final growth mean with *C. reinhardtii* (1113 ± 32 cells mL^{-1} on Day 7) (Table S3) compared to *E. coli* (1316.6 ± 7.64 cells mL^{-1} on Day 7) (Table S4), suggesting a preference for bacterial substrates. Similarly, *V. vermiformis* achieved a final growth mean of 997 ± 16 cells mL^{-1} with *C. reinhardtii* (Table S5) and 1076.6 ± 22.54 cells mL^{-1} with *E. coli* (Table S6), reflecting a comparable trend. Across all species, exponential growth was observed during the first three days, with *Acanthamoeba* genotype T3 demonstrating the steepest increase, reaching 1837 ± 47.52 cells mL^{-1} by Day 3 with *C. reinhardtii* at a dose of 50 μL (Figs. 2 and 3, Table S1). A similar pattern, accompanied by a higher growth mean, was observed when FLA were cultivated with *C. reinhardtii* at a dose of 100 μL (Fig. 1, Table S7, S8, S9). Notably, the growth of FLA on *C. reinhardtii* exhibited a strong positive correlation (Spearman, $r = 0.99$, $P < 0.0001$), further supporting the suitability of *C. reinhardtii* as a nutrient source. These findings emphasize that *C. reinhardtii* holds promise as an alternative to *E. coli* for FLA cultivation, particularly for *Acanthamoeba* genotype T3, which consistently demonstrated superior growth across experimental conditions. This observation has significant implications for sustainable laboratory practices and the development of algal-based cultivation techniques.

Cultivation of FLA on non-nutrient agar seeded with *S. obliquus*

The results demonstrate significant growth trends among different FLA types (*Acanthamoeba* genotype T3, *N. clarki*, and *V. vermiformis*) when cultivated with *S. obliquus* and *E. coli*. Overall, *Acanthamoeba* genotype T3 exhibited the highest growth compared to *N. clarki* and *V. vermiformis* across both food sources, with a final mean growth of 2423.3 ± 99.50 (Table S10) cells mL^{-1} on day 7 with *S. obliquus* (Friedman test; $P = 0.0012$), compared to 3088.3 ± 30.55 cells mL^{-1} (Table S11) with *E. coli* (Friedman test; $P = 0.0001$). Statistical analysis revealed that the food source significantly influenced growth, with *Acanthamoeba* genotype T3 showing higher early-stage growth with *E. coli* and *S. obliquus*. Similarly, *N. clarki* showed robust growth dynamics, achieving 1376.6 ± 25.17 cells mL^{-1} with *S. obliquus* (Table S12) and 1455 ± 32.78 cells mL^{-1} with *E. coli* on day 7 (Table S13), indicating its adaptability to different nutrient sources. However, *V. vermiformis* exhibited moderate growth, reaching 1160 ± 22.91 cells mL^{-1} with *S. obliquus* (Table S14) and 1076.6 ± 22.54 cells mL^{-1} with *E. coli* (Table S6), with no significant difference observed between food sources (Wilcoxon test, $P = 0.06$). The growth trajectories for all FLA with *S. obliquus* displayed a steady increase after day 3 (Figs. 3 and 4). FLA growth on *S.*

obliquus followed temporal patterns broadly comparable to those observed on *E. coli* (Spearman, $r = 0.99$, $P < 0.0001$) (Fig. 3, Table S15, S16, S17). These findings suggest that *S. obliquus* is a promising non-bacterial alternative to bacterial substrates for FLA cultivation. Its use may provide a more controlled cultivation platform for laboratory studies and may help reduce bacteria-associated complications in experimental investigations of FLA ecology and pathogenicity.

Cultivation of FLA on non-nutrient agar seeded with *C. vulgaris*

Figures 5 illustrates the growth patterns of three FLA types, *Acanthamoeba* genotype T3, *N. clarki*, and *V. vermiformis*, cultivated on two food sources, *C. vulgaris* and *E. coli*, at different concentrations. The data show that *Acanthamoeba* genotype T3 exhibited the highest growth rates across both food sources, reaching a peak of 1946.6 ± 35.12 cells mL^{-1} on *C. vulgaris* (Table S18) and 2171.6 ± 79.10 cells mL^{-1} on *E. coli* by Day 7 (Table S2, Fig. 5). *Naegleria clarki* displayed moderate growth, achieving higher growth on *E. coli* (mean 1316.6 ± 7.64 cells mL^{-1} on Day 7, Table S4) compared to *C. vulgaris* (946.6 ± 35.12 cells mL^{-1} on Day 7, Table S19). *V. vermiformis* showed the lowest overall growth but demonstrated a slightly better performance on *E. coli* (1076.6 ± 22.54 cells mL^{-1} on Day 7, Table S6, S20) compared to *C. vulgaris* (846.6 ± 30.55 cells mL^{-1} on Day 7, Table S21). Significant differences in growth rates among the three FLA types on *C. vulgaris* were observed (Friedman test, $P = 0.0001$). However, neither the food source (Mann–Whitney test, $P = 0.23$) nor the concentration (Mann–Whitney test, $P = 0.17$) showed significant effects on FLA growth (Table S22 & S23, S24). Nevertheless, the growth of FLA on *C. reinhardtii* and *S. obliquus* surpassed that on *C. vulgaris*, although the differences were not statistically significant (Kruskal–Wallis test; $P > 0.05$).

Discussion

The present study demonstrates that selected green algae can support the in vitro growth of potentially pathogenic free-living amoebae (FLA) on non-nutrient agar, thereby broadening the range of trophic substrates available for laboratory cultivation. Importantly, the significance of these findings lies not in suggesting that algal substrates universally outperform the conventional bacterial control, but rather in showing that algal-based systems can sustain measurable amoebal proliferation under controlled conditions. This distinction is critical, because the value of algal substrates is primarily methodological: they provide an alternative non-bacterial trophic platform for cultivating FLA in experimental settings

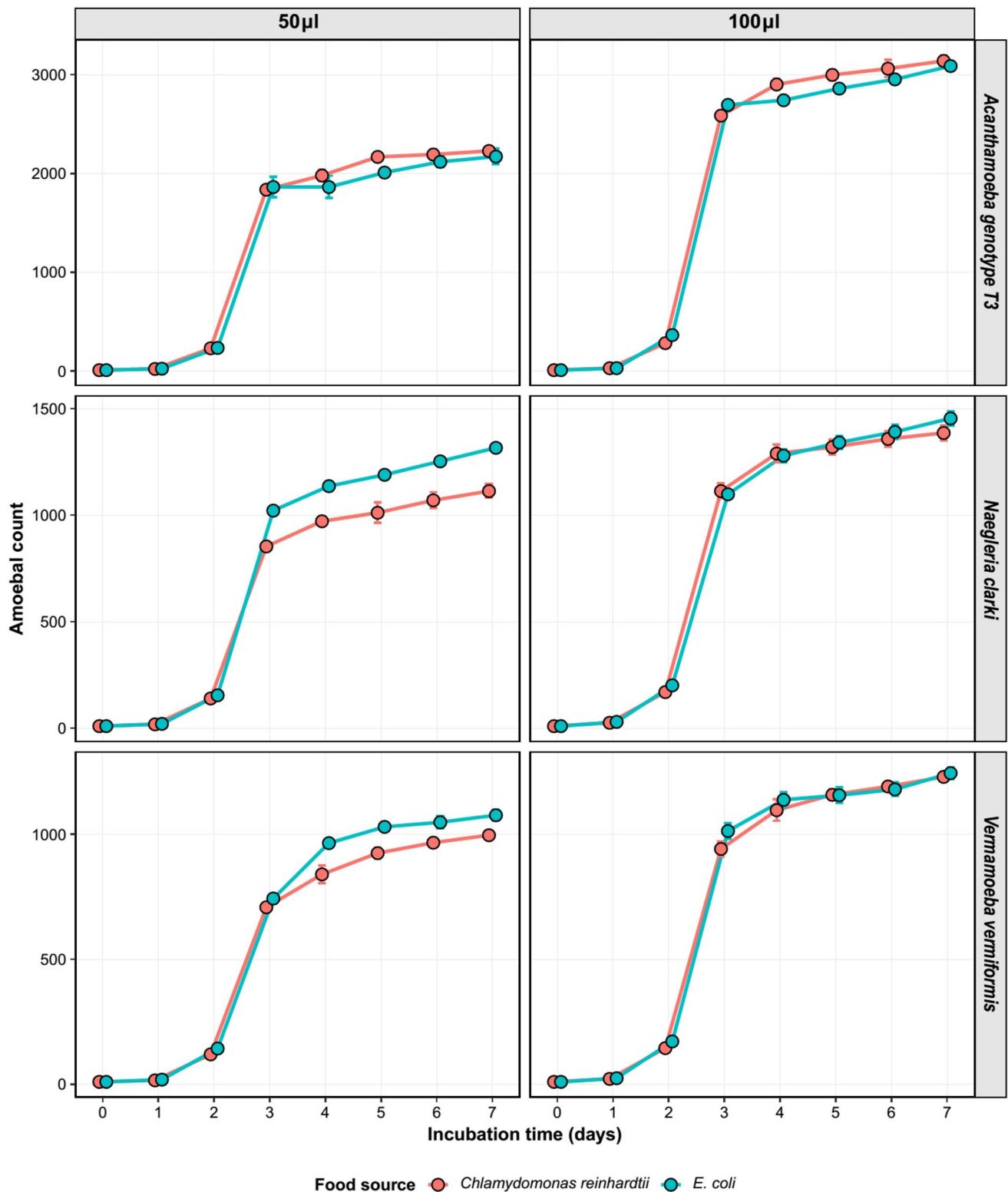


Fig. 2 Comparative growth dynamics of free-living amoebae on non-nutrient agar seeded with *Chlamydomonas reinhardtii* or *Escherichia coli* at 50 and 100 µL doses. Growth is expressed as amoebal counts, including trophozoites and cysts, per mL. Data are presented as the

mean $\pm$ standard deviation of three separate replicate wells ($n=3$) for each condition and sampling day. Each replicate well was enumerated once by one trained researcher

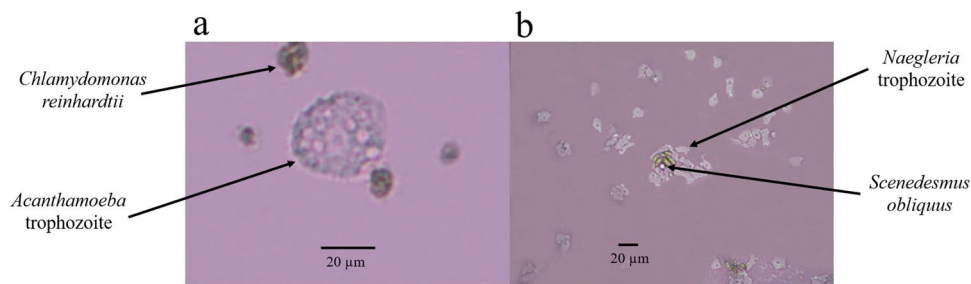


Fig. 3 Microscopic observation of free-living amoebae cultivated on non-nutrient agar seeded with algal food sources. The left panel shows an *Acanthamoeba* trophozoite in the presence of *Chlamydomonas reinhardtii*, while the right panel shows a *Naegleria*

trophozoite associated with *Scenedesmus obliquus*. Arrows indicate the amoeba trophozoites and corresponding algal cells used as alternative food sources

where bacterial prey may introduce additional biological complexity.

The use of bacteria-seeded non-nutrient agar, particularly with *E. coli*, remains the standard approach for cultivating FLA and has proven highly effective for isolation and routine maintenance (Khan 2006; Lorenzo-Morales et al. 2006). However, bacteria in amoebal culture systems may play roles that extend beyond nutrition alone. Many studies have shown that bacteria engulfed by amoebae can survive intracellular digestion, persist as endocytobionts, and exploit amoebae as protective environmental reservoirs (Greub and Raoult 2004; Thomas et al. 2010; Balczun and Scheid 2017). In particular, FLA have been described as “Trojan horses” that can shelter internalized microorganisms from adverse environmental conditions, disinfectants, antibiotics, and other biocidal agents, while also facilitating persistence, transmission, and adaptation of associated bacteria (Greub and Raoult 2004; Mungroo et al. 2021). Residence within amoebae may also promote bacterial interactions linked to virulence modulation and horizontal gene transfer, thereby complicating the interpretation of bacteria-fed cultivation systems in downstream ecological or pathogenicity studies (Greub and Raoult 2004; Thomas et al. 2010; Mungroo et al. 2021). Within this context, algal substrates are particularly attractive because they may reduce bacteria-associated experimental confounding while still supporting amoeba growth.

Among the tested amoebae, *Acanthamoeba* genotype T3 showed the strongest and most consistent proliferation across substrates, particularly on *S. obliquus* and *C. reinhardtii*. This pattern is biologically plausible given the broad trophic flexibility and ecological adaptability of protozoa, which is well known for exploiting diverse food particles and microbial resources in natural and experimental systems (Boenigk and Arndt 2002; Van Wichelen et al. 2010; Weger et al. 2024; Bilden et al. 2025). Amoebae do not merely ingest particles indiscriminately; rather, post-ingestion processing depends on prey properties such as size, digestibility, and structural composition (Bowers and

Olszewski 1983). Thus, the stronger growth observed on *S. obliquus* and *C. reinhardtii* compared with *C. vulgaris* may be associated with differences in cell wall properties among the tested algal species. The cell wall of *Chlorella* species has been reported to be particularly robust and recalcitrant to disruption (Voigt et al. 2014; Weber et al. 2022), which may reduce substrate accessibility to FLA under the experimental conditions tested. However, because cell wall digestibility and feeding mechanisms were not directly investigated in the present study, this interpretation should be considered a plausible hypothesis requiring further experimental validation.

In contrast to *Acanthamoeba* genotype T3, which consistently achieved the highest cell densities across all tested substrates, *N. clarki* and *V. vermiformis* exhibited lower overall growth irrespective of the food source. Nevertheless, both species showed measurable growth on all three green algal substrates, with growth levels that were generally comparable to those observed on the *E. coli* control. These findings suggest that the tested algal species can support the growth of multiple FLA taxa, although the magnitude of growth appears to be species dependent. Such interspecific differences are consistent with previous studies showing that amoebae growth responses can vary among taxa and may be influenced by both predator-specific characteristics and prey-related properties (Weekers et al. 1993; Pickup et al. 2007). The observed interspecific differences should not be interpreted solely on the basis of amoeba taxonomy. Previous evidence indicates that grazing and growth are often determined by specific interactions between individual amoeba and prey species or strains rather than by broad taxonomic affiliation alone (Weger et al. 2024).

It should also be noted that, although all three organisms are referred to as free-living amoebae because of their amoeboid morphology and mode of locomotion, they represent two distantly related evolutionary lineages: Amoebozoa (*Acanthamoeba* and *Vermamoeba*) and Discoba (*Naegleria*) (Adl et al. 2019). Accordingly, algal suitability should not

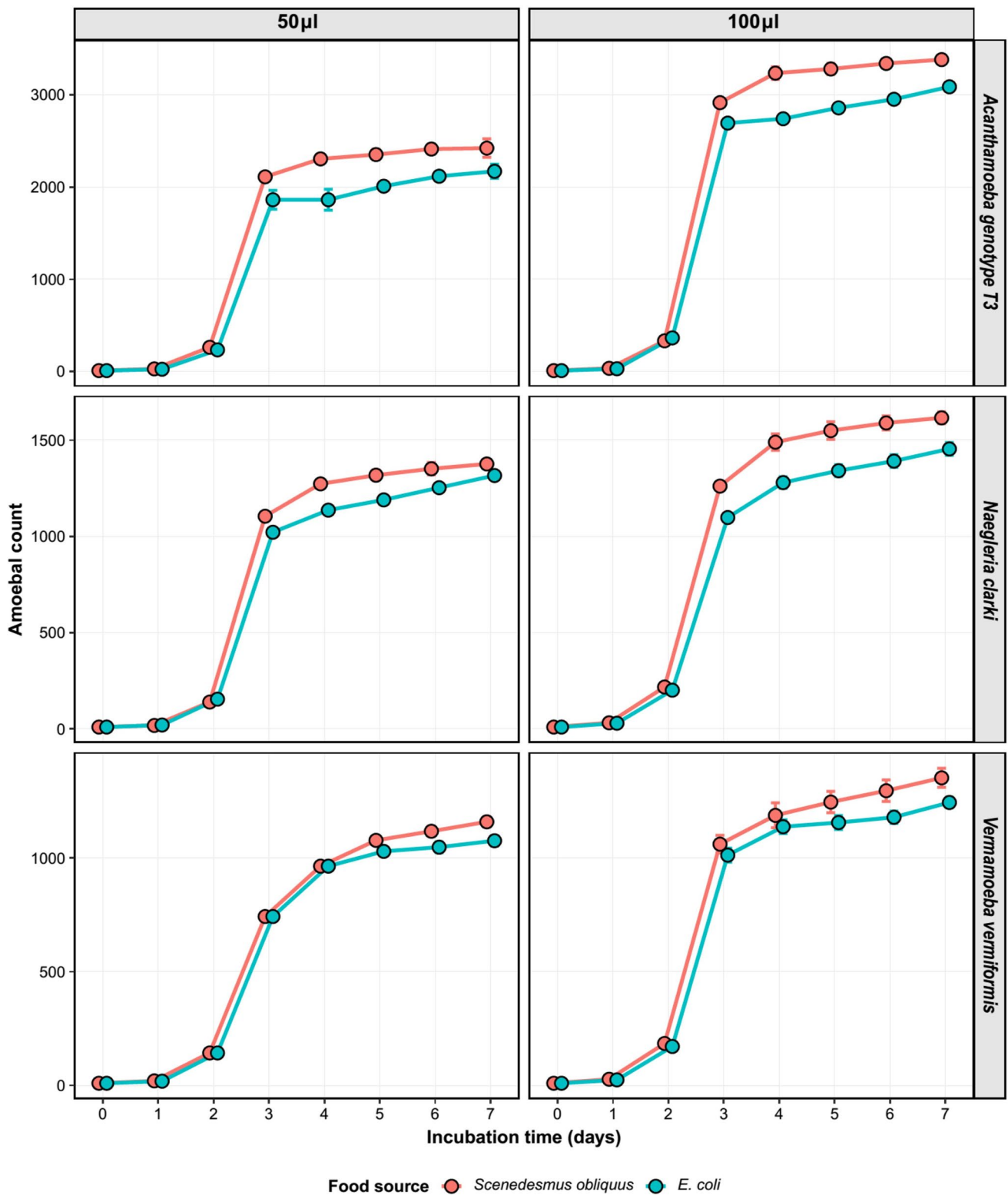


Fig. 4 Comparative growth dynamics of free-living amoebae on non-nutrient agar seeded with *Scenedesmus obliquus* or *Escherichia coli* at 50 and 100 µL doses. Growth is expressed as amoebal counts, including trophozoites and cysts, per mL. Data are presented as the

mean $\pm$ standard deviation of three separate replicate wells ($n=3$) for each condition and sampling day. Each replicate well was enumerated once by one trained researcher

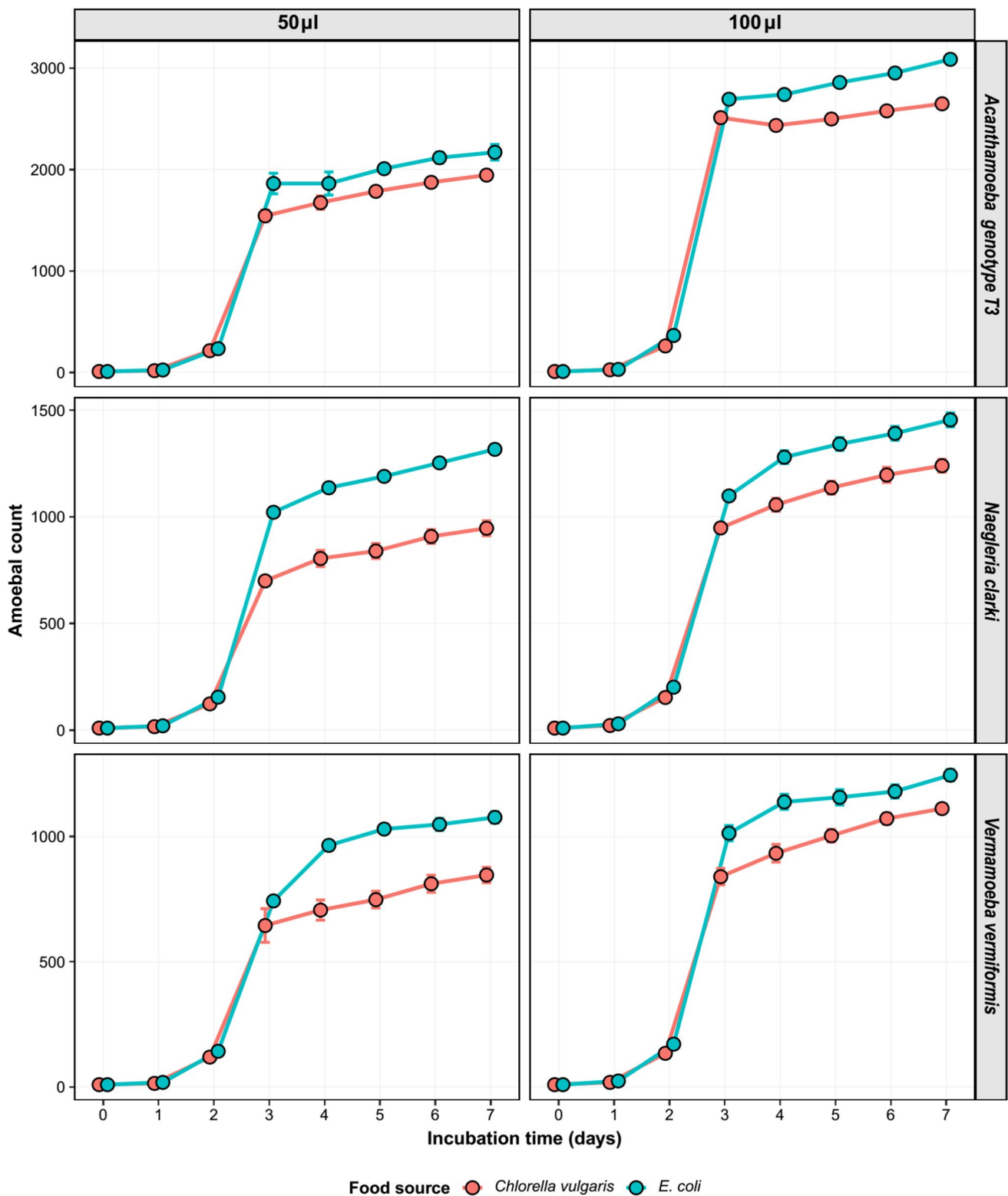


Fig. 5 Comparative growth dynamics of free-living amoebae on non-nutrient agar seeded with *Chlorella vulgaris* or *Escherichia coli* at 50 and 100 µL doses. Growth is expressed as amoebal counts, including trophozoites and cysts, per mL. Data are presented as the

mean $\pm$ standard deviation of three separate replicate wells ($n=3$) for each condition and sampling day. Each replicate well was enumerated once by one trained researcher

be interpreted as uniform across all amoebae, but rather as dependent on the compatibility between a given amoeba taxon and the structural and nutritional properties of the substrate. The positive correlations observed between growth patterns on algal and bacterial substrates further suggest that algal-supported proliferation broadly followed similar temporal trends to those obtained with *E. coli*, although such correlations should be interpreted cautiously and not as evidence of full nutritional equivalence.

The broader implication of these findings lies in the possibility of developing more controlled non-bacterial cultivation systems for FLA. In studies concerned with amoeba physiology, ecology, pathogenicity, or interactions with microorganisms, the use of bacterial prey may unintentionally create a biologically complex system in which intracellular bacterial survival and host–microbe interactions influence amoeba physiology and experimental outcomes (Matz et al. 2008; Thomas et al. 2010; Karnkowska et al. 2026). Previous work has shown, for example, that microorganisms surviving passage through protozoa may acquire enhanced tolerance to chlorination and other stresses, and may display altered phenotypic traits relevant to persistence and public health risk (Mattana et al. 2006; Mungroo et al. 2021). Against this background, the use of algae as a non-bacterial food source may offer a more controlled experimental platform for culturing amoebae while limiting the introduction of bacterial prey that could persist intracellularly.

These findings should be interpreted within the methodological scope of the study. Cell quantification was based on manual hemocytometer counting under microscopy, and although counts were performed independently by three skilled observers using standardized morphological criteria and triplicate measurements, some observer-dependent variability cannot be excluded, particularly at higher cell densities. The analyses were conducted in an ISO 17025-accredited laboratory operating under standardized procedures and external quality evaluation; however, formal statistical assessment of inter-observer agreement and blinded evaluation were not performed. In addition, inoculum standardization was based on cell counts rather than biomass, carbon content, or digestible nutritional value, and because algal and bacterial cells differ in size and biochemical composition, strict nutritional equivalence among substrates was not established. Accordingly, the results should be interpreted as demonstrating growth support and comparative response patterns rather than precise quantitative nutritional comparability. Although full axenicity of the algal cultures was not independently verified, the algal cells were washed three times with PAS prior to use, and the wash solution alone did not support FLA growth, suggesting that the observed proliferation was unlikely to result from the washing medium or loosely associated carryover, even though the presence of trace associated microorganisms cannot be excluded entirely.

Microscopic observations further indicated active ingestion of algal cells by the amoebae; however, ingestion was not quantified, and neither digestion efficiency nor algal depletion was measured. Likewise, physiological and molecular responses to the different substrates were not assessed. Taken together, these constraints indicate that the present study should be regarded as a proof-of-feasibility rather than a mechanistic analysis of amoeba algivory. Future work should therefore integrate rigorously controlled algal cultures, biomass-standardized feeding experiments, quantitative assessment of ingestion and substrate utilization, and physiological or transcriptomic approaches to clarify which algal traits most effectively support amoeba growth and to better evaluate the broader utility of algal-based cultivation systems.

Overall, the study shows that selected green algae, particularly *S. obliquus* and *C. reinhardtii*, can sustain measurable growth of potentially pathogenic FLA under laboratory conditions. These findings expand the methodological options available for amoeba cultivation and support the use of algal substrates as promising non-bacterial alternatives in experimental systems where bacterial prey may introduce undesirable ecological or interpretive complexity. Framed in this way, algal-based cultivation represents a useful and conceptually important direction for future research on FLA biology, ecology, and pathogenicity.

Conclusions

This study provides preliminary evidence that selected green algae can support the in vitro cultivation of potentially pathogenic free-living amoebae under controlled laboratory conditions. All tested amoeba species showed measurable growth on algal substrates, although the magnitude of the response varied among both amoebae and algal taxa. Among the examined strains, *Acanthamoeba* genotype T3 exhibited the most consistent and pronounced proliferation, particularly on *S. obliquus* and *C. reinhardtii*, indicating that algal suitability as a trophic substrate is species dependent rather than uniform across FLA. These findings highlight the potential of algal substrates to serve as promising non-bacterial alternatives for amoeba cultivation in experimental systems where bacterial prey may introduce additional ecological or interpretive complexity. At the same time, the observed growth patterns should not be interpreted as evidence of strict nutritional equivalence with conventional bacterial substrates, because the study was designed as a feasibility assessment rather than a mechanistic or biomass-normalized comparison. Overall, this work establishes a proof-of-concept foundation for algal-based cultivation of FLA and supports further investigation using rigorously

controlled algal cultures, quantitative feeding analyses, and physiological or molecular approaches to determine the broader applicability of this strategy in laboratory research on amoeba biology, ecology, and pathogenicity.

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Data availability Data will be made available on request.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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